

Robotics-for-Cacao Barter

Follow-up action item · Agroverse · 2026-08-06

Context. Three problems facing cacao farming: witches' broom disease, an aging farmer population with no labor to replace it, and inconsistent roasting. A robotics design expert partner is on board. Working-prototype reference: **Frasky**, the IIT vineyard robot (Reuters video, <https://youtu.be/hg8qYrjyYCU>). This follow-up turns that introduction into a concrete pilot.

1 · The three problems

Problem 1 — Witches' broom (*vassoura-de-bruxa*, *Moniliophthora perniciosa*)

The single highest-leverage canopy disease in Bahia cacao: infected brooms must be pruned out **and removed from the plantation** in dry season, or spores reinfect — cutting yields by 30–50% on affected trees. Detection and pruning are manual, repetitive, and easy to miss.

Problem 2 — Aging farmer population + labor shortage

Cacao farmers are aging, and there is **no younger generation stepping in** — farm management and harvest depend on manual labor that is increasingly hard to find and afford. The physical work (climbing, pruning, pod-cutting, carrying) is heavy, seasonal, and unattractive to new workers. The farm's know-how is concentrated in aging owners who cannot scale it.

Problem 3 — Roasting consistency

Roasting — the step that defines the flavor profile — is a **sequential decision process with delayed reward**: the operator acts on the roast curve (heat, airflow, drum speed, time), and only *after* the batch do you learn whether the flavor profile landed. Today that is skilled-craft, tuned batch-by-batch by experienced tasters — **not scalable** and hard to reproduce consistently across batches, roasters, and operators.

2 · Working prototype reference — Frasky (IIT)

Video (Reuters): <https://youtu.be/hg8qYrjyYCU> — "A hands-on robot tends grapes in Italy's vineyards"

- Built by the **Italian Institute of Technology + Bergamo-region agricultural stakeholders** (roboticist Dr. Manuel Catalano).
- Autonomously **monitors grape clusters** (camera + digital map of every cluster's location), **manipulates plants** (robotic arm), and **applies targeted spray treatments**.
- Designed for precision agriculture, **labor-shortage relief**, and sustainability — field-tested in vineyards (Reuters, Nov 2025).

Proof point: a permanent tree/vine crop's canopy work can be robotized. Cacao is the same category — this is the working prototype we point partners and farmers to.

Frasky capability (grape)	Cacao adaptation	Problems it addresses
Cluster monitoring via camera	Broom / pod detection via vision	P1 broom, P2 labor

Frasky capability (grape)	Cacao adaptation	Problems it addresses
Manipulate plants (robotic arm)	Broom pruning with cutter arm	P1 broom, P2 labor
Targeted spray treatments	Targeted biocontrol on brooms	P1 broom
—	Selective pod <i>harvest</i> (varied heights: trunk + branches)	P2 labor
—	Farm-management data (what to do, where, when)	P2 aging owners

3 · Roasting consistency as a reinforcement learning problem

The reframe — this is a reinforcement learning problem.

RL element	In the cacao roaster
State (observation)	Bean moisture, bean temperature / rate-of-rise, roast color, particle size distribution, volatile markers
Action	Heat input, airflow, drum speed, batch timing — the roast-curve decisions
Reward	Flavor-profile match (sensor + taste panel), batch-to-batch consistency, energy efficiency
Policy	A learned roast curve that reproduces a target flavor profile, batch after batch

The agent **learns the roast policy** over batches — each batch is a training episode, the reward comes at the end. That is exactly what makes it RL rather than a static recipe: the controller adapts to bean lot, moisture, and machine state to hit the flavor target.

The observation layer — particle / roast-state detection. The RL agent is only as good as what it can sense:

Detection approach	What it measures	Status
Machine vision (camera)	Bean color, surface browning, particle size distribution	■ proven in food lines
Hyperspectral / SWIR imaging	Moisture, roast degree, bean defects (slaty, moldy, under/over-fermented)	■ proven, conveyor-scalable
Portable NIR spectroscopy	Chemical markers of roast degree, in-line	■ proven, low-cost
Electronic nose + ANN	Volatile profile → predicted roast degree (94.4% accuracy reported)	■ proven

Why now / why us. Precedent exists (IMA Group's "AI Learning to Roast", model-based roast optimization) but is sold as expensive industrial software. A robotics + ML partner can build a **low-cost RL roast controller + particle detector** tuned to the beans and flavor profiles the Agroverse network actually runs —

turning roast flavor control from craft into a **scalable, self-improving process**.

4 · The proposal (barter model)

Offer robotics + automation services to Bahia/Pará farmers **in exchange for cacao as payment**:

Role	Who
Robot design & build	Robotics design expert (Gianluca)
Farmer network & deployment	Agroverse — Bahia operations (Ilhéus)
Market for cacao received	Agroverse (US import live)

The three problems → three workstreams:

Problem	Robotic/AI response	Feasibility
P1 · Witches' broom	Broom detection + pruning robot (à la Frasky)	■ tractable now
P2 · Aging farmers / labor	Harvest assist + farm-management data (what/where/when)	■ tractable (harvest ■■ harder R&D)
P3 · Roast consistency	RL roast controller + particle detection	■ parallel, lower-field-risk track

Honest MVP scoping. Broom *detection + pruning* (or monitoring + targeted treatment) is tractable now. *Selective cacao pod harvesting* is harder R&D — set expectations so the pilot stays deliverable.

5 · Pilot plan (1 season, measured)

1. **Scoping call with the robotics partner** — share the **Frasky video** (<https://youtu.be/hg8qYrjyYCU>) as the reference prototype; define MVP build, season timing, and cost basis; evaluate the roast-RL track (state sensors, reward signal, data capture per batch).
2. **Farmer pitch (PT-BR)** — value prop: labor shortage, aging-owner relief, broom yield loss, **no cash outlay**, paid in cacao at fair market BRL.
3. **2–3 test farms** in the Bahia (Ilhéus) network.
4. **Barter accounting** — value cacao received at market BRL/USD and record it into the supply ledger at fair market value.
5. **Measure, then scale** — % broom removed, harvest kg, labor-hours saved; roast batch-to-batch consistency. No scaling on unmeasured evidence.

6 · Next steps (owners)

- **Approve outreach copy** and introduce the robotics partner (WhatsApp).
- **Draft scoping-call question list** (incl. Frasky feasibility + roast-RL questions: observation sensors, reward definition, data per batch) and the PT-BR farmer pitch.

- **Robotics partner** — feasibility call; MVP definition.
- **Agroverse Brazil operations** — identify 2–3 test farms; coordinate season logistics.

7 · Tie-back to mission

Every broom pruned, every robot-assisted harvest, every aging farmer given a tool instead of a shovel — and every batch of cacao roasted to a reproducible flavor profile — keeps more trees producing and more **Amazon rainforest standing**. The 10,000-hectare rainforest regeneration goal is served by making cacao farming viable without clearing new land.

Draft v8 · pending approval before any outreach.